This article has been accepted for publication in IEEE Transactions on Control of Network Systems. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TCNS.2025.3638076

Sign-elevation angle rigidity for 3-D and almost global bearing-only formation stabilization in agents' local frames

Chinmay Garanayak and Dwaipayan Mukherjee

***Abstract*—In this paper, we propose sign-elevation angle rigidity and design bearing-only control laws in agents' local coordinate frames, without requiring any orientation synchronization or estimation algorithms. The theory of sign-elevation angle rigidity helps characterize unique formation shapes up to translation and rotation, given a set of elevation angle and volume constraints for 3-D frameworks. Moreover, we establish almost global stability results for formation control using elevation angle measurements obtained from bearing-only sensors in the agents' local frames. Recent elevation angle rigidity-based control laws guarantee only local stability and do not address flip, flex, and reflection ambiguities that can arise. Therefore, we first develop the theory of sign-elevation angle rigidity. Then, we propose control laws for single integrators and prove almost global stabilization to the desired formation shapes, using bearing-only measurements, when the desired formation is obtained by tetrahedral sub-formations in 3-D graphs. Control laws for non-holonomic agents are also analyzed. Finally, simulations are provided to validate the results.**

Index terms: bearing-only control, formation ambiguities, coordinate-free control, formation control

I. INTRODUCTION

Recent years have seen active research in distributed formation control of multi-agent systems, establishing it as an increasingly compelling area of study [1]–[4]. The central goal of formation control is to design decentralized control laws that enable agents to achieve and maintain a specific geometric configuration. A related challenge is in identifying geometric shapes using local constraints, which is addressed through rigidity theory and its extensions.

In position-based formation control, each agent is assigned a specific location and it must accurately detect its own position for the method to function effectively. Similarly, displacement-based formation control relies on relative displacement constraints between agents to define the desired formation shape [5]–[7]. This approach is closely linked to consensus protocols. Furthermore, displacement measurements are taken in a global frame, as the constraints are coordinate-dependent. In contrast, distance-based formation control defines the formation using inter-agent distance constraints. Distance rigidity theory addresses the question of determining unique formation shapes from such constraints [1], [8], [9]. Control laws developed under this framework require both distance and displacement information between agents but are coordinate-free [1], [3], [8], [10]–[12]. Works such as [13]–[15] have introduced formation shape definitions based on combined distance and angle constraints, giving rise to the concept of weak rigidity. The studies in [16], [17] proposed bearing rigidity theory, which enables the determination of unique formation shapes, up to translation and scale, using bearing constraints. In addition, [18] introduced bearing-ratio-of-distance rigidity theory, incorporating both bearing and ratio-of-distance constraints. However, the formation control laws in [17], [18] require inter-agent displacement measurements.

A. State of the Art

Vision-based applications can benefit from bearing measurements, which offer the advantage of utilizing low-cost passive optical sensors such as cameras. Bearing measurements also tend to exhibit less noise compared to displacement and distance measurements [19]. Additionally, they require a less complex sensory setup, reducing the payload on the agents. Consequently, bearing-only formation control has attracted significant research interest in recent years [4], [19], [20].

The primary goal of bearing-only formation control is to achieve a desired formation shape solely through bearing measurements, without relying on inter-agent distance or displacement data. In [4], bearing rigidity theory was employed to analyze bearing-only formation control for single integrators. Further developments in [20] presented tracking control strategies for single and double integrators, as well as unicycles, with constant-velocity leaders. Refs. [21], [22] investigated time-varying bearing-only formation control using the concept of persistence of excitation.

It is important to note that bearing measurements depend on a specific coordinate system, requiring agents to access a global frame when formation is specified via inter-agent bearing constraints. Therefore, to implement bearing-only control laws within local coordinate frames, orientation synchronization [23], [24] or orientation estimation algorithms [25], [26] are typically employed alongside the control law. However, such methods necessitate relative attitude measurements, which are often difficult to obtain in practice, or may require additional estimation algorithms based on local bearing sensing [27]. Refs. [28], [29] explored bearing-only formation control in agents' local frames using $SE(2)$ and $SE(3)$ rigidity theories but required additional controllable quantities to relate agents' local and global frames.

To eliminate the need for a global coordinate system, angle rigidity-based approaches were proposed for planar agents in [30], [31], incorporating signed angle constraints and establishing local exponential convergence to the desired formation. In [32], unsigned angle constraints were used to define the target shape. Refs. [33], [34] introduced triangular angle rigidity theory, specifying formation shapes via triangular angle constraints. Although almost global convergence

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was established, the control laws in [32], [34] still required relative displacement between agents. Ref. [35] addressed the specification of planar agents modeled as circular discs and proposed a bearing-only formation control scheme in 2D.

Elevation angle rigidity was explored in [36], where rods (in 2D) or balls (in 3D) were attached to agents to specify elevation angle constraints, describing the desired formation. A gradient-based bearing-only formation control law was then proposed in the local frame, guaranteeing local asymptotic stability. Since elevation angle constraints are coordinate-independent, the control laws in [36] did not require global coordinate information. In [37], elevation angle-based control laws were proposed for double integrators and non-holonomic agents. Ref. [38] extended this by designing control laws for single and double integrators under bounded disturbances. However, specifying only elevation angle constraints may lead to flip, flex, or reflection ambiguities in the final formation shape. These issues were not addressed in [36]–[38], where only local convergence was guaranteed.

Similar ambiguities are observed in distance-based formation control [39], [40]. In [39], such ambiguities were addressed using signed area constraints for 2D formations involving three or four agents. Ref. [40] introduced sign-rigidity theory, which used signed area and distance constraints to mitigate ambiguities in 2D graphs. Nevertheless, achieving almost global convergence and resolving formation ambiguities in elevation angle-based bearing-only control remains an open problem.

B. Contributions

In view of the above discussions, the main contributions of this work are as follows:

We introduce *sign-elevation angle rigidity* to characterize unique formation shapes (up to translation and rotation) using elevation angle and signed-volume constraints for graphs embedded in 3D space. Further, we derive the *sign-elevation angle rigidity matrix* and establish a rank condition for sign-elevation angle rigidity. We also prove almost global convergence to the desired formation using only local bearings, for 3D formations constructed via a *signed-elevation-Henneberg* method. Stability is analyzed for both single-integrators and non-holonomic agents. Compared to existing results, this work offers the following distinctive features:

(i) Compared to the bearing-only control laws in [4], [20]–[22], our approach does not require agents to know their orientation relative to a global frame. Unlike [23]–[25], no orientation synchronization or estimation algorithms are needed. Additionally, unlike [28], [29], our method requires no orientation control as an extra constraint.

(ii) In contrast with angle rigidity-based bearing-only control laws in [30], [31], we use elevation angle constraints, eliminating scale ambiguity in the final formation shape. Consequently, our formations behave as rigid bodies, able to preserve inter-agent distances under perturbations, whereas formations in [30], [31] exhibit translational, rotational, and scale invariance, and may thus shrink or expand under the effects of disturbances.

(iii) Unlike [36], [38], which rely solely on elevation angle constraints, our framework combines elevation angles with signed-volume constraints to specify 3D formations, eliminating flip, flex, and ordering ambiguities. Furthermore, while prior works [30], [31], [35], [36], [38] only established local stability, we prove almost global convergence. Hence, agents can reach the desired shape from almost any initial configuration. This is a novel contribution in the context of bearing-only formation control using local frames.

(iv) Contrary to [34], [41], which require inter-agent displacement measurements, our control law relies solely on local bearing information. In [42], global stability was achieved through a localization-based controller where agents' desired and actual positions were estimated from angle constraints and angle measurements, respectively. Also [42] required absolute position data for the leaders. In contrast, our method is implementable directly via vision sensors, with no need for global position estimates.

(v) Compared with [40], which used distance and signed-area constraints for planar (2D) frameworks, our approach handles 3D formations using elevation angle and signed-volume constraints, allowing broader and more practical applications. Moreover, while [40] employed inter-agent displacement in control, our method uses bearing-only measurements, enabling implementation with vision-only sensors. The structure of the Hessian matrix in our formulation also differs significantly from that in [40].

Some preliminary results, on local stability, using sign-elevation angle rigidity were presented in [43]. In this article, we establish almost global stability, consider non-holonomic agents in the control design, and present a detailed discussion on sign-elevation angle rigidity along with various proofs that were not given in [43]. The distinctive features of the current work, compared with existing literature, are summarized in Table I.

Notations: $\|\cdot\|$ is used for 2-norm, and $\|\cdot\|_1, \|\cdot\|_\infty$ represents 1-norm, and ∞-norm, respectively. $|\cdot|$ represents modulus operation. For $x = [x_1 \;\; x_2 \;\; x_3]^\top \in \mathbb{R}^3, [x]^\times := \begin{bmatrix} 0 & -x_3 & x_2 \\ x_3 & 0 & -x_1 \\ -x_2 & x_1 & 0 \end{bmatrix} \in \mathbb{R}^{3\times 3}$. Symbols $\hat{i}, \hat{j}, \hat{k}$ represent the orthogonal unit vectors along the x, y, z-axes of the coordinate system, respectively, and $\times$ denotes the cross product. $\langle \cdot, \cdot \rangle$ is the inner product operation. $\text{Diag}\{a_1, \ldots, a_n\} \in \mathbb{R}^{n\times n}$ is a diagonal matrix, where $a_1, \ldots, a_n$ are its diagonal entries. $J_1, J_2, J_3 \in \mathbb{R}^{3\times 3}$ correspond to rotations of $\pi/2$ along the principal axes in $\mathbb{R}^3$. $\text{sign}(\cdot)$ is the signum function, i.e., for $x \in \mathbb{R}$, $\text{sign}(x) = 0, -1, 1$ for $x = 0, x < 0$, and $x > 0$, respectively. For $x \in \mathbb{R}^n$, we have $\text{sign}(x) := [\text{sign}(x_1) \; \ldots \; \text{sign}(x_n)]^\top$.

II. PRELIMINARIES

A. Graph theory

This subsection presents some notions from graph theory [44]. We model the interaction among agents by an undirected graph, $\mathcal{G}(\mathcal{V}, \mathcal{E})$, where $\mathcal{V} = \{1, \ldots, n\}$ is the node-set, and the

TABLE I
COMPARISON WITH STATE OF THE ART

References	Knowledge of global frame or absolute orientation	Use of orientation synchronization or estimation	Type of measurement	Stability result	Agent dynamics:*	Applicability for 2-D or 3-D
[4]	yes	no	bearings	almost global	(a)	2-D or 3-D
[20]	yes	no	bearings	almost global	(a),(b),(c)	2-D or 3-D
[21], [22]	yes	no	bearings	almost global	(a),(b)	2-D or 3-D
[23], [24]	no	yes	bearings	almost global	(b)	2-D or 3-D
[26]	no	yes	bearings	almost global	(b),(c)	2-D or 3-D
[25]	no	yes	bearings	almost global	(a)	2-D or 3-D
[30], [31], [35], [36]	no	no	bearings	local	(a)	2-D or 3-D
[38]	no	no	bearings	local	(a),(b)	2-D or 3-D
[34], [41]	yes	no	relative displacement	global	(a)	2-D or 3-D
[40]	no	no	relative displacement	almost global	(a),(c)	2-D
present paper	no	no	bearings	almost global	(a), (c)	3-D

* (a) $\Rightarrow$ Single integrator, (b) $\Rightarrow$ Double integrator, (c) $\Rightarrow$ Non-holonomic agent

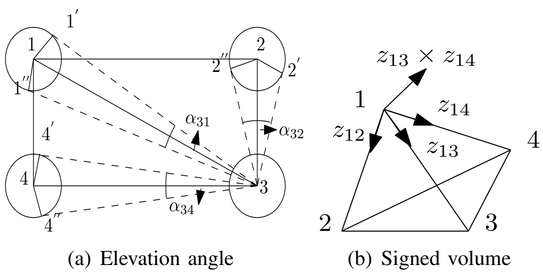

Fig. 1. (a) Elevation angle measurements in 3-D, (b) Signed volume $V_{1234} = r_c^{-3} z_{12}^\top (z_{13}\times z_{14}) = f_{12}f_{13}f_{14}b_{12}^\top(b_{13}\times b_{14})$

set of edges is denoted by $\mathcal{E}$, which is a subset of the Cartesian product of $\mathcal{V}$ with itself. The neighbor set of the i-th agent is denoted by $\mathcal{N}_i$, with n_i representing the number of neighbors. The positions of all the agents in $\mathbb{R}^d$ are combined to form a *realization*, denoted by $p := [p_1^\top,\dots,p_n^\top]^\top \in \mathbb{R}^{3n}$.

B. Agent dynamics

We consider the following systems for control design.

1) Single integrators: Single integrator agents are given by

$$\dot{p}_i = u_i \;\; i\in\mathcal{V}, \tag{1}$$

where $u_i \in \mathbb{R}^3$ is the control input.

2) Non-holonomic agents: These are modeled as

$$\dot{p}_i = h_i\bar{v}_i,\;\; \dot{h}_i = w_i \times h_i \;\; i\in\mathcal{V}, \tag{2}$$

where $h_i \in \mathbb{R}^3$ is the heading direction of agent i, $\bar{v}_i \in \mathbb{R}$ is the speed along the heading direction, and $w_i \in \mathbb{R}^3$ is the angular velocity. The control inputs are $\bar{v}_i$ and w_i.

C. Elevation angle rigidity

Let $z_{ij} := p_j - p_i, d_{ij} := ||z_{ij}||$, and $b_{ij} := \frac{z_{ij}}{||z_{ij}||} = \frac{z_{ij}}{d_{ij}}$, be the relative displacement, distance, and the bearing vector between agents i and j, respectively. Let r_c be the radius of the ball attached to each agent. The points $j^{'}$ and $j^{''}$ are obtained by drawing tangents from agent i to the ball attached with agent j such that $i, j, j^{'}, j^{''}$ lies on the same plane. For 3-D case the elevation angle is defined as [36] $\angle j^{'} i j^{''} = \alpha_{ij} := \arccos(b_{ij'}^\top b_{ij''}) = 2\angle j^{'} ij = 2\arcsin(\frac{r_c}{d_{ij}}) \in (0, \frac{\pi}{3})$. Fig. 1 (a) shows elevation $\alpha_{31}, \alpha_{32}, \alpha_{34}$ for 3-D case. The elevation angle function is defined as

$$E(p) : \mathbb{R}^{dn} \to \mathbb{R}^{m_e},\;\; E(p) := [f_1,\dots,f_{m_e}]^\top,$$

where $f_k := \operatorname{cosec}(\frac{\alpha_k}{2}) = \operatorname{cosec}(\frac{\alpha_{ij}}{2}) = \frac{d_{ij}}{r_c}$, $k := (i,j) \in \mathcal{E}$, and $f_k \in (2,\infty)$ for 3-D. Let, m_e be the number of elevation angle constraints, i.e., $m_e := |\mathcal{E}|$. When elevation angle constraints define the desired formation,

$$f_k = f_k^*,\;\; k := (i,j)\in\mathcal{E} \implies E = E^*, \tag{3}$$

i.e., the desired formation, in terms of elevation angle constraints, is defined through a desired realization of the elevation angle function. Note that a double subscript, such as f_{ij}, will be used to describe neighboring nodes/agents involved. A single subscript, like f_k, can also be used where k is the edge index. The derivative of the elevation angle constraints yields

$$\frac{df_{ij}}{dt} = r_c^{-1} b_{ij}^\top (v_j - v_i) \text{ (for 3-D)}, \tag{4}$$

where $v_i,\ i\in\mathcal{V}$, is the velocity of agent i. The elevation angle rigidity matrix is defined by: $R_e := \frac{\partial E}{\partial p} \in \mathbb{R}^{m_e \times 3n}$.

III. MAIN RESULTS

We first present sign-elevation angle rigidity theory. Then, we design suitable control laws for single integrators and non-holonomic agents and prove almost global stability.

If the desired formation is specified only with elevation angle constraints, an ambiguous final shape may result.

Definition 1. *(Ambiguous configuration) A formation shape with a configuration $p^{'} := [(p_1^{'})^\top \dots (p_n^{'})^\top]^\top \in \mathbb{R}^{3n}$ is an ambiguous configuration, if $E(p^{'}) = E^*$, where $p^{'}$ cannot be obtained by translation or rotation of the desired configuration p^*, i.e., $p_i^{'} \neq Rp_i^* + T,\ i\in\mathcal{V}$, where $R\in\mathbb{R}^{3\times3}$ and $T\in\mathbb{R}^3$ represent rotations and translations in $\mathbb{R}^3$.*

Hence, some agents may flip their positions, satisfying the same elevation angle constraints, thereby giving rise to an ambiguous formation shape. More details on formation ambiguities can be found in [43]. For 3-D, to avoid such

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ambiguous configurations, we introduce the signed-volume constraints given by

$$V_{ijkl} = r_c^{-3} z_{ij}^\top (z_{ik} \times z_{il}) = V_{ijkl}^* \ (i,j),(i,k),(i,l) \in \mathcal{E},$$
$$\implies V_{ijkl} = f_{ij} f_{ik} f_{il} b_{ij}^\top (b_{ik} \times b_{il}) = V_{ijkl}^*. \quad (5)$$

Fig 1(b) illustrates a signed-volume constraint $V_{1234} = r_c^{-3} z_{12}^\top (z_{13} \times z_{14}) = f_{12} f_{13} f_{14} b_{12}^\top (b_{13} \times b_{14})$. The directions of different vectors involved in it are also shown in Fig 1(b). We may observe from (5) that if any of the agents from i, j, k, l flips its position, maintaining the inter-agent elevation angle constraints, the sign (positive or negative) of the volume will change. This is because $b_{ij}^\top (b_{ik} \times b_{il})$ will change its sign if any of the agents flips. Hence, the nomenclature signed-volume is used. The multiplication of r_c^{-3} in (5) will help in obviating the knowledge of r_c in the control law. To specify a desired formation, a set of elevation angle constraints, along with a set of signed-volume constraints, is used. All the signed-volume constraints are stacked in $V := [V_1, \ldots, V_{m_v}]^\top \in \mathbb{R}^{m_v}$. Here, $m_v := |V|$, i.e., m_v represents the number of volume constraints. The subscripts for the volume constraints may be expressed as a single variable or as four variables (when the nodes involved are specified).

A. Sign-elevation angle rigidity in 3-D

In this subsection, sign-elevation angle rigidity is developed in 3-D. Note that for $x, y \in \mathbb{R}^3$, $x \times y = [x]^\times y$.

Definition 2. *(Framework) Define the volume index set by $\mathcal{A} = \{(i,j,k,l) \in \mathcal{V}^4 | V_{ijkl} \in V\}$. Now, a* framework *associated with $\mathcal{G}$ and p is denoted as $\mathcal{F}(\mathcal{G}, p, \mathcal{A})$, where p represents a realization.*

Remark 1. *The ordering of the nodes matters while defining the volume index set, as it is generated by the edges to characterize the signed-volume. Hence, $(i,j,k,l) \in \mathcal{A} \nRightarrow (j,k,l,i) \in \mathcal{A}$, for 3-D.*

Definition 3. *(Equivalence of formations) Two frameworks $\mathcal{F}_1(\mathcal{G}, p, \mathcal{A})$ and $\mathcal{F}_2(\mathcal{G}, q, \mathcal{A})$ in 3-D over the same graph $\mathcal{G}(\mathcal{V}, \mathcal{E})$, and volume index set $\mathcal{A}$, are equivalent if*

$$\alpha_{ij}(p_i, p_j) = \alpha_{ij}(q_i, q_j) \ \forall (i,j) \in \mathcal{E},$$
$$V_{ijkl}(p_i, p_j, p_k, p_l) = V_{ijkl}(q_i, q_j, q_k, q_l) \ \forall (i,j,k,l) \in \mathcal{A}.$$

Definition 4. *(Congruence of formations) Two frameworks $\mathcal{F}_1(\mathcal{G}, p, \mathcal{A})$ and $\mathcal{F}_2(\mathcal{G}, q, \mathcal{A})$ in 3-D, over the same graph $\mathcal{G}(\mathcal{V}, \mathcal{E})$, and volume index set $\mathcal{A}$, are congruent if*

$$\alpha_{ij}(p_i, p_j) = \alpha_{ij}(q_i, q_j) \ i,j \in \mathcal{V},$$
$$V_{ijkl}(p_i, p_j, p_k, p_l) = V_{ijkl}(q_i, q_j, q_k, q_l) \ i,j,k,l \in \mathcal{V}.$$

Definition 5. *(Sign-elevation angle rigidity) A framework $\mathcal{F}(\mathcal{G}, p, \mathcal{A})$ is sign-elevation angle rigid if there exists an $\epsilon > 0$ such that, all frameworks $\mathcal{F}'(\mathcal{G}, q, \mathcal{A})$ in 3-D satisfying $||p - q|| < \epsilon$ are congruent to it.*

Definition 6. *(Global sign-elevation angle rigidity) A framework, $\mathcal{F}(\mathcal{G}, p, \mathcal{A})$, is globally sign-elevation angle rigid if any other framework $\mathcal{F}'(\mathcal{G}, q, \mathcal{A})$ that is equivalent to $\mathcal{F}(\mathcal{G}, p, \mathcal{A})$, is congruent to it.*

Next, we introduce the notion of infinitesimal sign-elevation angle rigidity. For this, we first define the following sign-elevation angle rigidity function: $F_e^v : \mathcal{X} \to \mathbb{R}^{(m_e + m_v)}$, where $\mathcal{X} \subset \mathbb{R}^{3n}$:

$$F_e^v(p) := \ [f_1 \ldots, f_{m_e}, V_1, \ldots, V_{m_v}]^\top, \quad (6)$$

where $E = [f_1, \ldots, f_{m_e}]^\top \in \mathbb{R}^{m_e}$ and $V = [V_1, \ldots, V_{m_v}]^\top \in \mathbb{R}^{m_v}$ are the vectors containing the elevation angle constraints and the signed volume constraints, respectively. The number of elevation angle constraints and signed volume constraints are m_e and m_v, respectively.

Remark 2. *From Definition 5, it is clear that a framework, $\mathcal{F}(\mathcal{G}, p, \mathcal{A})$, is sign-elevation angle rigid if there exists a neighborhood, $\mathcal{U}_p$, around p such that $(F_e^v)^{-1}(F_e^v(p)) \cap \mathcal{U}_p = (F^v_{\frac{n(n-1)}{2}})^{-1}(F^v_{\frac{n(n-1)}{2}}(p)) \cap \mathcal{U}_p$, where $F^v_{\frac{n(n-1)}{2}}$ is the sign-elevation angle rigidity function for a complete graph.*

Now, we define the infinitesimally sign-elevation angle rigid motions of a framework. These are motions of the framework $\mathcal{F}$ on a differentiable path, $\varrho(t)$, such that the sign-elevation angle rigidity function, i.e., F_e^v, remains constant, i.e., $\dot{F}_e^v \equiv 0$. We say that infinitesimal motions of the entire framework ($\dot{p}$) are *trivial* if they correspond to only translations and rotations of the desired formation shape, i.e., for *trivial* motions we have $\dot{F}^v_{\frac{n(n-1)}{2}} \equiv 0$, where $F^v_{\frac{n(n-1)}{2}}$ is the sign-elevation rigidity function associated with a complete graph. Equivalently, the trivial motions of the framework are given by

$$p_i(t) = R(t) p_i(t_0) + T(t), \ i \in \mathcal{V}, t > 0,$$

where $R(t) \in \mathbb{R}^{3\times 3}$ and $T(t) \in \mathbb{R}^3$ represent rotation and translation motions in $\mathbb{R}^3$. The infinitesimal sign-elevation angle rigidity is introduced to characterize all the trivial motions of the framework.

Next, we introduce the sign-elevation angle rigidity matrix, denoted as R_e^v, which is defined as the Jacobian of the sign-elevation angle rigidity function, i.e.,

$$R_e^v := \ \frac{\partial F_e^v}{\partial p} = \begin{bmatrix} \frac{\partial E}{\partial p} \\ \\ \frac{\partial V}{\partial p} \end{bmatrix} = \begin{bmatrix} R_e \\ \\ R_v \end{bmatrix} \in \mathbb{R}^{(m_e + m_v) \times 3n}.$$

Taking the derivative of the volume constraint (5) gives,

$$\begin{aligned} &\frac{dV_{ijkl}}{dt} = r_c^{-3} z_{ij}^\top (z_{ik} \times (v_l - v_i)) - r_c^{-3} z_{ij}^\top (z_{il} \times (v_k - v_i)) \\ &+ (z_{ik} \times z_{il})^\top (v_j - v_i) \\ &= r_c^{-1} (f_{ij} f_{ik} b_{ij}^\top ([b_{ik}]^\times (v_l - v_i) - f_{ij} f_{il} b_{ij}^\top ([(b_{il}]^\times (v_k - v_i)) \\ &- f_{ik} f_{il} b_{il}^\top ([b_{ik}]^\times (v_j - v_i)), \end{aligned} \quad (7)$$

where v_i, v_j, v_k, v_l are velocities of agents i, j, k, l, respectively. Combining (3) and (5), we get

$$\dot{F}_e^v = \frac{\partial F_e^v(p)}{\partial p} \dot{p} = \begin{bmatrix} \frac{\partial E}{\partial p} \\ \\ \frac{\partial V}{\partial p} \end{bmatrix} \dot{p} = 0. \quad (8)$$

Definition 7. *(Infinitesimal sign-elevation angle rigidity) We say a framework is infinitesimally sign-elevation angle rigid if all of its infinitesimal motions are trivial. Also, note that trivial*

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motions belong to the kernel of the sign-elevation rigidity matrix associated with a complete graph. Hence, a framework $\mathcal{F}(\mathcal{G}, p, \mathcal{A})$ is infinitesimally sign-elevation angle rigid if Null$\left(\frac{\partial F_e^v(p)}{\partial p}\right)$=Null$\left(\frac{\partial F(p)^v_{\frac{n(n-1)}{2}}}{\partial p}\right)$, where $F(p)^v_{\frac{n(n-1)}{2}}$ is the sign-elevation rigidity function associated with a complete graph.

The following Lemma and the subsequent theorem provide a way of checking the infinitesimal rigidity of a framework.

Lemma 1. *For the sign-elevation angle rigidity matrix (R_e^v), we have $L_p := \text{span}\{1_n \otimes I_3, (I_n \otimes J_1)p, (I_n \otimes J_2)p, (I_n \otimes J_3)p\} \subseteq$ Null(R_e^v), and Rank(R_e^v) $\leq 3n-6$.*

Proof. From [36, Lemma 1] it is known that $\text{span}\{1_n \otimes I_2, (I_n \otimes J_1)p, (I_n \otimes J_2)p, (I_n \otimes J_3)p\} \subseteq \text{Null}(R_e)$. As $R_e^v = \begin{bmatrix} R_e \\ R_v \end{bmatrix}$, we need to show that $\text{span}\{1_n \otimes I_2, (I_n \otimes J_1)p, (I_n \otimes J_2)p, (I_n \otimes J_3)p\} \subseteq \text{Null}(R_v)$. Define the new edge set as $\mathcal{E}' := \mathcal{E} \cup \mathcal{E}_v$, where $\mathcal{E}_v = \{(i,j),(i,k),(i,l) | (i,j,k,l) \in \mathcal{A}\}$. Let, $m_t = |\mathcal{E}'|$. With the new edge set, the induced graph is denoted by $\mathcal{G}' = (\mathcal{V}, \mathcal{E}')$. We have $R_v = \frac{\partial V}{\partial p} = \frac{\partial V}{\partial z'}\frac{\partial z'}{\partial p} = \frac{\partial V}{\partial z'}\bar{H}'$, where $z' = [z_1^\top, \ldots, z_{m_t}^\top]^\top$, $z_k \in \mathcal{E}'$, and $\bar{H}' = (H' \otimes I_3)$, with H' being the incidence matrix of graph $\mathcal{G}'$. Now, $\text{span}\{1_n \otimes I_3\} \subseteq \text{Null}(\bar{H}' \otimes I_3)$, implies $\text{span}\{1_n \otimes I_3\} \subseteq R_e^v$. Similarly it can be shown easily $(I_n \otimes J_k) \subseteq R_v$, $k \in 1,2,3$ by considering the l-th element in V, i.e., $V_l = r_c^{-3} z_p^\top (z_q \times z_r)$, where $l \in \{1, \ldots, m_v\}$, and $p, q, r \in \{1, \ldots, m_t\}$, and showing $\frac{\partial V_l}{\partial p}(I_n \otimes J_k)p = 0$, $k = \{1,2,3\}$. This completes the proof. □

Theorem 1. *In 3-D, a framework $(\mathcal{G}, p, \mathcal{A})$ is infinitesimally sign-elevation angle rigid if and only if rank(R_e^v)$= 3n-6$.*

Proof. From Lemma 1 it is clear that rank of R_e^v is less than or equal to $3n-6$, and $\text{span}\{1_n \otimes I_3, (I_n \otimes J_1)p, (I_n \otimes J_2)p, (I_n \otimes J_3)p\} \subseteq \text{Null}(R_v^e)$. Observe that $1_n \otimes I_3$ corresponds to rigid body translations in 3-D (i.e. translations along $\hat{i}$, $\hat{j}$, and $\hat{k}$), and $(I_n \otimes J_\beta)p$, $\beta = 1,2,3$ correspond to rigid body rotations, which are the trivial infinitesimal rigid body motions in 3-D. Hence, clearly $\text{rank}(R_e^v) = 3n-6$ if and only if these trivial motions satisfy eqn. (8). Thus, according to Definition 7, framework $(\mathcal{G}, p, \mathcal{A})$ is infinitesimally sign-elevation angle rigid. □

Theorem 2. *A framework $\mathcal{F}(\mathcal{G}, p, \mathcal{A})$ is infinitesimally sign-elevation angle rigid if and only if it is sign-elevation angle rigid.*

Proof. Define sets $\mathcal{S}(p) := (F_e^v)^{-1}(F_e^v(p))$ and $\mathcal{K}(p) := (F^v_{\frac{n(n-1)}{2}})^{-1}(F^v_{\frac{n(n-1)}{2}}(p))$. Hence, the set $\mathcal{S}(p)$ contains all configurations $p' \in \mathbb{R}^{3d}$ such that the framework $\mathcal{F}(\mathcal{G}, p', \mathcal{A})$ is equivalent to $\mathcal{F}(\mathcal{G}, p, \mathcal{A})$, and the set $\mathcal{K}(p)$ contains all configurations $p' \in \mathbb{R}^{3d}$ such that the framework $\mathcal{F}(\mathcal{G}, p', \mathcal{A})$ is congruent to $\mathcal{F}(\mathcal{G}, p, \mathcal{A})$. Now, let us consider a framework $\mathcal{F}(\mathcal{G}, p, \mathcal{A})$ that is not sign-elevation angle rigid. Then there exists at least a configuration p' in the neighborhood of p, i.e., $\mathcal{U}(p)$, such that $p' \in \mathcal{S}(p) \backslash \mathcal{K}(p)$. Therefore, the framework $\mathcal{F}(\mathcal{G}, p, \mathcal{A})$ is not infinitesimally sign-elevation angle rigid according to Definition 7. Hence, by contrapositive, we have proved the statement in one direction.

To prove the other side, assume that the framework $\mathcal{F}(\mathcal{G}, p, \mathcal{A})$ is sign-elevation angle rigid. Now, if it is not infinitesimally sign-elevation angle rigid, then there will be continuous deformation p' from p such that $p' \in \mathcal{S}(p) \backslash \mathcal{K}(p)$ implying a contradiction with respect to Remark 2. This completes the proof. □

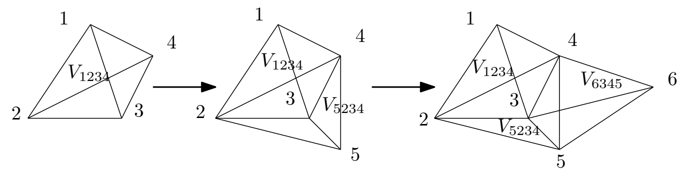


Fig. 2. Example of sign-elevation-Henneberg construction method for generating globally sign-elevation angle rigid frameworks in 3-D.

1) Sign-elevation-Henneberg construction in 3-D: (Constructing globally sign-elevation angle rigid frameworks:) The following steps are followed to generate global sign elevation angle rigid frameworks in 3-D. Here, new frameworks are generated by adding tetrahedral frameworks with one new signed-volume constraint. First, we assume the initial framework has $|\mathcal{V}| = 4, |\mathcal{E}| = 6$, and $\mathcal{A} = 1$. For a given globally sign elevation angle rigid framework $\mathcal{F}(\mathcal{G}, p, \mathcal{A})$, an agent v is added to it such that it has three more elevation angle constraints and one signed-volume constraint. Now, the new framework $\bar{\mathcal{F}}(\bar{\mathcal{G}}, \bar{p}, \bar{\mathcal{A}})$ consists of tetrahedral frameworks and $\bar{\mathcal{G}} = (\bar{\mathcal{V}}, \bar{\mathcal{E}}), \bar{\mathcal{V}} = \mathcal{V} \cup \{v\}, \bar{\mathcal{E}} = \mathcal{E} \cup \{(i,v),(j,v),(k,v)\}, \bar{\mathcal{A}} = \mathcal{A} \cup \{(\bar{i}, \bar{j}, \bar{k}, \bar{l})\}$, and $\bar{p} = [p^\top \; p_v^\top]^\top \in \mathbb{R}^{3(n+1)}$, where $i,j,k \in \mathcal{V}$, with $(i,j),(j,k),(k,i) \in \mathcal{E}, \bar{i}, \bar{j}, \bar{k}, \bar{l} \in \{i,j,k,v\}$ and $\bar{i} \neq \bar{j} \neq \bar{k} \neq \bar{l}$. Fig. 2 demonstrates these steps. It is important to note that here we cannot add the new edges to any arbitrary nodes $i,j,k \in \mathcal{V}$, but need to add the edges such that $(i,j),(j,k),(k,i) \in \mathcal{E}$ is satisfied, e.g., in Fig. 2 node 6 is added with edges to $3,4,5$; however, the addition of agent 6 to nodes $1,3,5$ is not allowed as $(1,5) \notin \mathcal{E}$. The reason behind such a construction will be clarified later.

Remark 3. *The name Sign-elevation-Henneberg construction is inspired by the Henneberg construction used for generating infinitesimal distance rigid frameworks [1] with only distance constraints, whereas here we have used elevation angle constraints along with signed-volume constraints. Note that, Henneberg construction is used for generating infinitesimal distance rigid frameworks and not globally distance rigid frameworks. The proposed sign-elevation-Henneberg construction can however generate globally sign-elevation rigid frameworks, which is proved in the following proposition.*

Proposition 1. *An infinitesimally sign-elevation angle rigid framework $\mathcal{F}(\mathcal{G}, p, \mathcal{A})$ in 3-D is globally signed elevation angle rigid if it is generated by sign-elevation Henneberg construction.*

Proof. As the framework, $\mathcal{F}(\mathcal{G}, p, \mathcal{A})$, is infinitesimally sign-elevation angle rigid, from Theorem 1 we have $\text{rank}(R_e^v) = 3n-6$. Hence, from Lemma 1, R_e^v is of maximal row rank,

This article has been accepted for publication in IEEE Transactions on Control of Network Systems. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TCNS.2025.3638076

and using [8, Proposition 2] there exists a neighborhood $\mathcal{U}_p$ such that $(F_e^s)^{-1}(F_e^s(p)) \cap \mathcal{U}_p$ is a 6-dimensional smooth manifold which corresponds to 3-D rigid transformations. Further, from Definition 5, it follows that $\mathcal{F}(\mathcal{G}, p, \mathcal{A})$ is sign-elevation angle rigid. In addition, using the sign-elevation Henneberg construction, we can uniquely determine the position of each agent, as the position of each agent in a sequence is uniquely determined by 3 elevation angle and 1 signed-volume constraints. Hence, from Definition 6, we conclude that $\mathcal{F}(\mathcal{G}, p, \mathcal{A})$ is globally sign-elevation angle rigid. □

Definition 8. *(Strong distance rigidity in 3-D) A distance rigid framework $\mathcal{F}(\mathcal{G}, p)$ in 3-D is called strongly distance rigid if we have $(p_i - p_j)$,$(p_i - p_k)$, and $(p_i - p_l)$ independent for all $(i,j), (i,k), (i,l) \in \mathcal{E}$. Physically, this means the vectors $(p_i - p_j)$,$(p_i - p_k)$, and $(p_i - p_l) \in \mathbb{R}^3$ do not lie on the same plane.*

Proposition 2. *If a framework, $\mathcal{F}(\mathcal{G}, p)$, is strongly distance rigid in $\mathbb{R}^3$, then the signed framework $\mathcal{F}_1(\mathcal{G}_1, p, \mathcal{A})$ where $\mathcal{G}_1 = \mathcal{G}$, (i.e., sensing topology for distances in distance rigid graph $\mathcal{G}$ is same as elevation angle constraints in $\mathcal{G}_1$), is infinitesimally sign-elevation angle rigid in $\mathbb{R}^3$.*

Proof. Consider the signed framework $\mathcal{F}_1(\mathcal{G}_1, p, \mathcal{A})$ and replace its elevation angle constraints with distance constraints to obtain $\mathcal{G}$. As we have $\mathcal{F}(\mathcal{G}, p)$ strongly distance rigid in 3-D, using similar induction-based arguments in [11, Proposition 1], it can be proved that $\mathcal{F}(\mathcal{G}, p)$ is infinitesimal distance rigid. Now, a framework is elevation angle rigid if and only if it is distance rigid [36, Proposition 2]. Hence, the framework with elevation angle constraints $\mathcal{F}(\mathcal{G}_1, p)$ is elevation angle rigid. So, there exists a non zero $(3n-6) \times (3n-6)$ minor of R_e [36]. Now, from the definition of sign-elevation angle rigidity matrix $R_e^v = \begin{bmatrix} R_e \\ R_v \end{bmatrix}$ there exists a $(3n-6) \times (3n-6)$ non zero minor for R_e^v. Then, the proof follows from Theorem 1. □

B. Bearing-only formation control using sign-elevation angle rigidity for n agents

This section presents a bearing-only control law to achieve the desired formation specified by elevation angle and signed-volume constraints, considering n single integrators. We have the following signed volume and desired volume constraints for 3-D:

$$\bar{V} = \kappa[\dots, V_{ijkl}, \dots]^\top \in \mathbb{R}^{m_v},$$
$$\bar{V}^* = \kappa[\dots, V_{ijkl}^*, \dots]^\top \in \mathbb{R}^{m_v},$$

where $(i,j,k,l) \in \mathcal{A}$, and $\kappa > 0$ is a positive constant. The formation error is defined by

$$e(p) = [E(p)^\top \ \bar{V}^\top]^\top - [E^{*\top} \ (\bar{V}^*)^\top]^\top. \tag{9}$$

Let $u = [u_1^\top, \dots, u_n^\top]^\top$ contain the control inputs of all agents. The following gradient-based control law is proposed for the single integrators:

$$\dot{p} = u = -r_c \nabla\Phi = -r_c (\bar{R}_e^v(p))^\top e(p), \tag{10}$$

where $\Phi = \frac{1}{2} e^\top e$, and $\bar{R}_e^v = [R_e^\top \ \ \kappa R_v^\top]^\top$, which we call the modified sign-elevation angle rigidity matrix. Although κ is multiplied with the part corresponding to the signed volume of the sign-elevation angle matrix, observe that Null$(\bar{R}_e^v)$ =Null(R_e^v), hence the rank of R_e^v is the same as that of $\bar{R}_e^v$. From (10), the control law for i-th agent is:

$$\dot{p}_i = u_i = -r_c (\bar{R}_e^v)^\top (i,:)(F_e^v - (F_e^v)^*) \quad \text{for 3 D}, \tag{11}$$

where $(\bar{R}_e^v(:,i))^\top$ is the i-th row of $(\bar{R}_e^v)^\top$. Using (4) and (7), the i-th column of $\bar{R}_e^v$ is given by:

$$\begin{aligned}\bar{R}_e^v(:,i) = [&0, \dots, -r_c^{-1} b_{ij}^\top, 0, \dots, -r_c^{-1} b_{ik}^\top, 0 \dots, -r_c^{-1} b_{il}^\top, \\ &0, \dots, 0, -\kappa r_c^{-1} f_{ij} f_{ik} b_{ij}^\top [b_{ik}]^\times + \kappa r_c^{-1} f_{ij} f_{il} b_{ij}^\top [b_{il}]^\times + \\ &\kappa r_c^{-1} f_{ik} f_{il} b_{il}^\top [b_{ik}]^\times, \dots, 0]^\top,\end{aligned} \tag{12}$$

where $(i,j), (i,k), (i,l) \in \mathcal{E}$, $(i,j,k,l) \in \mathcal{A}$. Here, $f_{ij} := \text{cosec}(\frac{\alpha_{ij}}{2}) = \frac{1}{\sin(\frac{\alpha_{ij}}{2})} = \frac{1}{\sqrt{(1-\cos(\alpha_{ij}))/2}} = \frac{1}{\sqrt{(1-b_{ij'}^\top b_{ij''})/2}}$. Also, signed-volume constraints can be represented as $V_{ijkl} := r_c^{-3} z_{ij}^\top (z_{ik} \times z_{il}) = f_{ij} f_{ik} f_{il} b_{ij}^\top (b_{ik} \times b_{il})$. Hence, the control law u_i in 3-D also requires bearing-only information. Also, additional information on r_c is not required, which is clear from (10) and (12).

Remark 4. *Observe that for the control design, we have modified the signed-volume constraints using a positive gain κ. The error function and the potential function are defined with these constraints. Since the rank of the modified sign-elevation angle matrix ($\bar{R}_e^v$) is the same as that of R_e^v, the introduction of κ does not affect the rigidity properties of the framework. However, κ will play a crucial role in proving almost global stability with the gradient control law* (10).

Theorem 3. *Suppose $\mathcal{Q}$ denotes a set of p corresponding to formations of single integrators (described by* (1)*) that are congruent to a desired infinitesimally sign-elevation angle rigid formation. If the initial position $p(0) \in \mathcal{B}_{p^*}$, where $\mathcal{B}_{p^*}$ is a neighborhood of p^*, and $p^* \in \mathcal{Q}$, using the control law* (10)*, $p(0)$ locally exponentially converges to a desired formation $p^{des} \in \mathcal{Q}$.*

Proof. This can be proved using arguments based on center manifold theorem [12], [45], and is hence omitted. □

From Theorem 3, it is clear that under the control law (10), the desired configuration is a stable equilibrium point. However, it only establishes local convergence results. In the subsequent subsection, we establish almost global convergence for the frameworks generated by a sign-elevation Henneberg construction with large κ, implying that for sufficiently large κ, we cannot have other stable equilibria.

Remark 5. *We adopted a sign-elevation Henneberg construction to build a globally sign-elevation angle rigid framework. However, this alone does not guarantee almost global convergence to the desired configuration under control law* (10).

Now, we state the following Lemma, which will be helpful in our subsequent analyses.

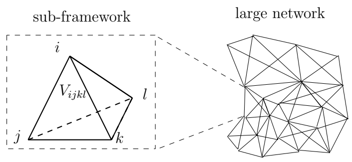


Fig. 3. Examples of sub-frameworks $\mathcal{F}_w(\mathcal{G}_w, \beta_w, \mathcal{A}_w)$ for 3-D case for $w \in \{1, 2, \ldots, \psi\}$

Lemma 2. *Let $\beta_j \in \mathbb{R}^{12}$ be a regular point*[1] *for $F^v_{e_j}$ $j \in \{1, 2, \ldots, \psi\}$, then there exists a derivative map D_{ζ_i} of a map $\zeta_i(\beta_i)|_{\mathcal{G}_j}$, for β_i is in the neighborhood of β_j, i.e., $\beta_i \in \mathcal{B}_{\beta_j}$, close to $\mathcal{F}_j(\mathcal{G}_j, \beta_j, \mathcal{A}_j)$ such that*

$$D_{\zeta_i}(v_i)|_{\mathcal{G}_j} = \begin{cases} v_i \in \mathbb{R}^{d|\mathcal{V}_i|} & \text{if } i = j \\ v_i \in T_{\zeta_i(\beta_i)}(F^v_{e_j})^{-1} F^v_{e_j}(\beta_j) & \text{otherwise,} \end{cases}$$

where $T_x\mathcal{M}$ is the tangent space to a manifold $\mathcal{M}$ at point x.

Proof. Refer to Appendix A for the proof. □

Remark 6. *From Lemma 1 and Theorem 1, we have β_j, $j \in \{1, 2, \ldots, \psi\}$ is a regular point of $F^v_{e_j}$ implies the sub-frameworks are sign-elevation angle rigid. Hence, from Lemma 2 and [8, Proposition 2] $(F^v_{e_j})^{-1} F^v_{e_j}(\beta_j)), j \in \{1, 2, \ldots, \psi\}$ are the motions of $\mathcal{F}_j(\mathcal{G}_j, \beta_j, \mathcal{A}_j)$ in $SE(3)$.*

C. Almost global stabilization for single integrators in 3-D

In this subsection, we establish almost global convergence in 3-D, using the gradient control law in (10) for sufficiently large values of κ, when the desired framework is constructed using the sign-elevation Henneberg construction.

As we have a sign-elevation-Henneberg construction, we may split the signed framework, $\mathcal{F}(\mathcal{G}, p, \mathcal{A})$, into ψ sub frameworks. Here, $\mathcal{F}(\mathcal{G}, p, \mathcal{A})$ is divided into $\mathcal{F}_w(\mathcal{G}_w, \beta_w, \mathcal{A}_w)$, $w \in \{1, 2, \ldots, \psi\}$. Each partition, $\mathcal{F}_w(\mathcal{G}_w, \beta_w, \mathcal{A}_w)$, consists of four agents, and hence for the sub-graph $\mathcal{G}_w = (\mathcal{V}_w, \mathcal{E}_w)$ we have $\mathcal{V}_w = \{i, j, k, l\}$ and $\beta_w = [p_i^\top \ p_j^\top \ p_k^\top \ p_l^\top]^\top \in \mathbb{R}^{12}$, $i, j, k, l \in \mathcal{V}_w$ is a realization associated with the sub-framework. Hence, $p = [\beta_1^\top \ \ldots \ \beta_\psi^\top]^\top \in \mathbb{R}^{3n}$. Also, we have $\mathcal{E} = \bigcup_{w=1}^{\psi} \mathcal{E}_w$ and $\mathcal{E}_w \cap \mathcal{E}_{w'} = \{(i, j), (j, k), (k, i)\}$, when $w, w' \in \{1, 2, \ldots, \psi\}, w \neq w'$, and $|\mathcal{E}_w| = 6$. Also, $\mathcal{A} = \bigcup_{w=1}^{\psi} \mathcal{A}_w$ with $\mathcal{A}_w \cap \mathcal{A}_{w'} = \varnothing$ when $w, w' \in \{1, 2, \ldots, \psi\}$ and $w \neq w'$ with $|\mathcal{A}_w| = 1$. It follows that $\mathcal{V} = \bigcup_{w=1}^{\psi} \mathcal{V}_w$ and $|\mathcal{V}_w| = 4$, and if β_w and $\beta_{w'}$ are adjacent for $w, w' \in \{1, 2, \ldots, \psi\}, w \neq w'$, then $\mathcal{V}_w \cap \mathcal{V}_{w'} \neq \varnothing$, otherwise $\mathcal{V}_w \cap \mathcal{V}_{w'} = \varnothing$. Examples of sub-frameworks are given in Fig. 3.

The following functions correspond to different sub-frameworks. The vectors containing the actual and the desired elevation angle constraints are given by

$$E_w(\beta_w) = [\ldots \ f_{ij} \ \ldots]^\top \in \mathbb{R}^{|\mathcal{E}_w|}, \tag{13}$$
$$E^*_w(\beta^*_w) = [\ldots \ f^*_{ij} \ \ldots]^\top \in \mathbb{R}^{|\mathcal{E}_w|}, \tag{14}$$

where $w \in \{1, 2, \ldots, \psi\}$ and $(i, j) \in \mathcal{E}_w$. The actual and desired signed-volume constraint for a sub-framework are

$$V_w(\beta_w) = \kappa V_{ijkl} \in \mathbb{R}, V^*_w = \kappa V^*_{ijkl} \in \mathbb{R}, \tag{15}$$

where $w \in \{1, 2, \ldots, \psi\}$ and $(i, j, k, l) \in \mathcal{A}_w$. The sign-elevation angle rigidity matrix is

$$\bar{R}^v_{e_w} = \frac{\partial F_w}{\partial \beta_w} \in \mathbb{R}^{(|\mathcal{E}_w|+1)\times 12}, \ w \in \{1, 2, \ldots, \psi\}, \tag{16}$$

where $F_w := [E_w^\top(\beta_w) \ \ V_l(\beta_w)]^\top \in \mathbb{R}^{|\mathcal{E}_w|+1}$ is the sign-elevation angle rigidity function associated with a sub-framework, $\mathcal{F}_w(\mathcal{G}_w, \beta_w, \mathcal{A}_w)$. Hence, the formation error associated with each framework is

$$e_w(\beta_w) = [E_w^\top(\beta_w) \ \ V_w(\beta_w)]^\top - [E_w^{*\top} \ V^*_w]^\top. \tag{17}$$

Now, we modify the edge errors associated with the sub-frameworks such that the new error is divided by the number of sub-frameworks connected with the edge, i.e., $\hat{e}_{e_{ij}} := \frac{1}{\sqrt{\varrho_{ij}}} e_{e_{ij}}$, where $e_{e_{ij}} := f_{ij} - f^*_{ij}$, and ϱ_{ij} is the number of sub-frameworks associated with edge (i, j). According to this definition, from Fig. 2, we have $\hat{e}_{e_{34}} := \frac{1}{\sqrt{3}} e_{e_{34}}$, i.e., $\varrho_{34} = 3$, and $\hat{e}_{e_{12}} := e_{e_{12}}$, i.e., $\varrho_{12} = 1$. Thus, we have the modified formation error associated with each sub-framework $w \in \{1, 2, \ldots, \psi\}$ given by

$$\hat{e}_w(\beta_w) = K e_w(\beta_w) \in \mathbb{R}^{|\mathcal{E}_w|+1}, \tag{18}$$

where $K := \text{Diag}\{\ldots, \frac{1}{\sqrt{\varrho_{ij}}} \ldots, 1\} \in \mathbb{R}^{(|\mathcal{E}_w|+1)\times(|\mathcal{E}_w|+1)}$, $(i, j) \in \mathcal{E}_w$. The potential function for the sub-framework is $\Phi_w = \frac{1}{2}||\hat{e}_w(\beta_w)||^2$, $w \in \{1, 2, \ldots, \psi\}$ and its Hessian is

$$\mathcal{H}_{\beta_w} = \frac{\partial^2 \Phi_w}{\partial \beta_w^2} \in \mathbb{R}^{12\times 12}. \tag{19}$$

Using this modified definition of edge errors, we have $\Phi = \sum_{w=1}^{\psi} \Phi_w$.

Lemma 3. *Consider a system of n single integrators and a framework $\mathcal{F}(\mathcal{G}, p, \mathcal{A})$, generated by a sign-elevation-Henneberg construction in 3-D. If $p = p^*$ is a configuration corresponding to a stable equilibrium under the control law* (10), *then with sufficiently large $\kappa > 0$, the following hold:*
(i) At equilibrium, the signed volume has the same sign as the desired volume, i.e., the sign of the equilibrium volume must match with the sign of the desired volume.
(ii) At equilibrium, the framework $\mathcal{F}(\mathcal{G}, p^*, \mathcal{A})$, must be infinitesimally sign-elevation angle rigid.

Proof. Let us calculate the Hessian of Φ, which is also the negative Jacobian matrix of (10):

$$\mathcal{H}_p = \frac{\partial^2 \Phi}{\partial p^2} \in \mathbb{R}^{3n\times 3n}. \tag{20}$$

The Hessian associated with each sub-framework is given by:

$$\mathcal{H}_{\beta_l} = (R^v_{e_w})^\top K^4 R^v_{e_w} + \bar{N}_2 + \bar{N}_1. \tag{21}$$

The exact expression of matrices $R^v_{e_w}, \bar{N}_2, \bar{N}_1$, and the Hessian $\mathcal{H}_{\beta_l}$ can be found in Appendix B. Let $p^\dagger$ be a realization with an incorrect sign than that of the desired signed volume. This means there exists a signed-volume V_{ijkl} such that at

[1] [8] A realization β_j is a regular point of a function $F^v_{e_j}$ if $\text{rank}\left(\frac{\partial F^v_{e_j}}{\partial \beta_j}\right) = \max\left\{\text{rank}\left(\frac{\partial F^v_{e_j}}{\partial \beta_j}\right) | \beta_j \in \mathbb{R}^{d|\mathcal{V}_j|}\right\}$

equilibrium, $V_{ijkl} \leq 0$, while $V^*_{ijkl} > 0, (i,j,k,l) \in \mathcal{A}$. Observe that for this sub-framework $\mathcal{F}_w(\mathcal{G}_w, \beta_w, \mathcal{A}_w),\ l \in \{1,2,\dots,\psi\}$ we will have $V_{ijkl} - V^*_{ijkl} \neq 0, (i,j,k,l) \in \mathcal{A}_w$ at $p^\dagger$. Now, we prove that when this happens, we have a negative eigenvalue for $\mathcal{H}_p$ at $p^\dagger$ for sufficiently large κ, making $p = p^\dagger$ an unstable equilibrium.

We use a contradiction-based argument. Assume that for sufficiently large κ, $p = p^\dagger$ is a stable equilibrium. As $\bar{R}^v_{e_w} = [R^\top_{e_w}\ \ \kappa R^\top_{v_w}]^\top$, where $R_{e_w} = \frac{\partial F_{e_w}}{\partial \beta_w}$ and $R_{v_w} = \frac{\partial V_{ijkl}}{\partial \beta_w}$ for $(i,j,k,l) \in \mathcal{A}_w, w \in \{1,2,\dots,\psi\}$, using (21) we may write

$$\mathcal{H}_{\beta_w} = R^\top_{e_w} \bar{K} R_{e_w} + \kappa^2 R^\top_{v_w} R_{v_w} + \bar{N}_1 + \bar{N}_2,$$

where $\bar{K} := \text{Diag}\{\dots, \frac{1}{\varrho_{ij}{}^2}, \dots\} \in \mathbb{R}^{|\mathcal{E}_w| \times |\mathcal{E}_w|}, (i,j) \in \mathcal{E}_w$. Define $y := [y_1^\top\ \epsilon y_2^\top\ \mathbf{0}^\top\ \mathbf{0}^\top]^\top \in \mathbb{R}^{12}$, with $y_1 \in \mathbb{R}^3$ being a unit vector perpendicular to $(z_{kl} \times z_{kj})$ where $(i,j,k,l) \in \mathcal{A}_w$ and $w \in \{1,2,\dots,\psi\}$, $y_2 \in \mathbb{R}^3$ is a non-zero vector with $\epsilon > 0$ being a small positive number. Using the fact that $\frac{\partial V_{ijk}}{\partial \beta_l} = r_c^{-3}[(z_{kl} \times z_{kj})^\top\ (z_{ik} \times z_{il})^\top\ (z_{il} \times z_{ij})^\top\ (z_{ij} \times z_{ik})^\top], (i,j,k,l) \in \mathcal{A}_w, w \in \{1,2,\dots,\psi\}$, we have

$$\begin{aligned} & y^\top(\kappa^2 R^\top_{v_w} R_{v_w} + \bar{N}_2) y && (22)\\ &= \kappa^2 \underbrace{\epsilon^2 y_2^\top (z_{ik} \times z_{il})(z_{ik} \times z_{il})^\top y_2}_{=y^\top R^\top_{v_w} R_{v_w} y} + \underbrace{\kappa \epsilon y_1^\top \Delta y_2 e_{v_{ijkl}}}_{=y^\top(\bar{N}_2) y}, && (23)\end{aligned}$$

where $e_{v_{ijkl}} = \kappa(V_{ijkl} - V^*_{ijkl})$, and $\Delta := -[z_{kl}]^\times + [z_{ik}]^\times - [z_{il}]^\times = -2[z_{kl}]^\times \in \mathbb{R}^{3\times 3}$ (because $[z_{ik}]^\times - [z_{il}]^\times = -[z_{kl}]^\times$). Observe (from Appendix (B)) that these two terms $R^\top_{e_w} \bar{K} R_{e_w}$ and $\bar{N}_1$ are independent of κ, hence

$$y^\top \mathcal{H}_{\beta_w} y = \kappa^2(\epsilon y_1^\top \Delta y_2 (V_{ijkl} - V^*_{ijkl}) + O(\epsilon^2) + O(\bar{\epsilon}^2)),$$

where $\bar{\epsilon} = \frac{1}{\kappa}$. We may thus can always choose y so that $y_1^\top \Delta y_2 (V_{ijkl} - V^*_{ijkl})$ is negative. This is because due to our assumption $(V_{ijkl} - V^*_{ijkl}) < 0$ and $y_1^\top \Delta y_2 = \langle y_2, \Delta^\top y_1 \rangle$, we have to choose y_2 such that it makes an acute angle with the vector $\Delta^\top y_1$. Also, note that $\Delta^\top y_1 = 2[z_{kl}]^\times y_1 = 2(z_{kl} \times y_1)$, and hence, $\Delta^\top y_1$ is always a non-zero vector. Hence, for sufficiently large κ we will have $y^\top \mathcal{H}_{\beta_l} y$ is negative at $p^\dagger$. So, for sufficiently large κ, when an incorrect sign for $(i,j,k,l) \in \mathcal{A}_w$ occurs, $\mathcal{H}_{\beta_w}$ has a negative eigenvalue. We have $\Phi = \sum_{w=1}^{\psi} \Phi_w$, which implies $\mathcal{H}_p = \sum_{w=1}^{\psi} \hat{\mathcal{H}}_{\beta_w}$, where $\hat{\mathcal{H}} = \frac{\partial^2 \Phi_w}{\partial p^2} \in \mathbb{R}^{3n \times 3n}$. Hence $\hat{\mathcal{H}}_{\beta_w}$ is constructed by padding zero rows and columns to $\mathcal{H}_{\beta_w}$. This implies that there exists $\hat{y} \in \mathbb{R}^{3n}$, which can be constructed by adding zeros to y such that $\hat{y}^\top \hat{\mathcal{H}}_{\beta_w} \hat{y} < 0$, and $\hat{\mathcal{H}}_{\beta_w}$ has at least one negative eigenvalue at $p^\dagger$. Also, note that the sub-framework $\mathcal{F}_l(\mathcal{G}_l, \beta_l, \mathcal{A}_l),\ l \in \{1,2,\dots,\psi\}$ has at least one agent common with the adjacent sub-framework $\mathcal{F}_l'(\mathcal{G}_l', \beta_l', \mathcal{A}_l'),\ l' \in \{1,2,\dots,\psi\}\backslash\{l\}$. Hence, $\mathcal{H}_p$ is not a block diagonal matrix consisting of $\mathcal{H}_{\beta_l}, l \in \{1,2,\dots,\psi\}$ as the diagonal blocks. Hence, we cannot directly conclude that $\mathcal{H}_p$ has a negative eigenvalue at $p^\dagger$, from the fact that $\hat{\mathcal{H}}_{\beta_l}$ has a negative eigenvalue at $p^\dagger$. However, now we invoke Lemma 2 to reach this conclusion. From the above analysis, it is clear that at equilibrium for a sufficiently large κ each of the sub-frameworks $\mathcal{F}_l(\mathcal{G}_l, \beta_l, \mathcal{A}_l),\ l \in \{1,2,\dots,\psi\}$, is an infinitesimally sign-elevation angle rigid configuration. Hence, from Lemma 1 and Theorem 1, $\bar{R}^v_{e_l}$ is of maximum row rank, which implies $\beta_l, l \in \{1,2,\dots,\psi\}$ is a regular point for $F^s_{e_l}$. Also, we have span$(L_p) \subset$ Null$(\hat{\mathcal{H}}_{\beta_l})$ [11, Lemma 2], [46]. Hence, it follows from Lemma 2 that there exists a vector $D_{\zeta_l}(\hat{y})|_{\mathcal{G}_l}$ such that

$$(D_{\zeta_l}(\hat{y})|_{\mathcal{G}_{l'}})^\top \hat{\mathcal{H}}_{\beta_l}(p^\dagger) D_{\zeta_l}(\hat{y})|_{\mathcal{G}_{l'}} = \begin{cases} \theta < 0 \text{ if } l = l' \\ 0,\ l \neq l'. \end{cases} \quad (24)$$

Now, using the fact that $\mathcal{H} = \sum_{l=1}^{\psi} \hat{\mathcal{H}}_{\beta_l}$, we have $(D_{\zeta_l}(\hat{y})|_{\mathcal{G}_{l'}})^\top \mathcal{H}_p(p^\dagger) D_{\zeta_l}(\hat{y})|_{\mathcal{G}_{l'}} = \theta < 0$, which is a contradiction. Hence, we conclude that for sufficiently large $\kappa > 0$, we cannot have an incorrect signed volume at the equilibrium, i.e., we will have $V_{ijkl} > 0$ if $V^*_{ijk} > 0$, and hence part (i) is proved. Further, note that this means we cannot have $V_{ijkl} = 0$ at the equilibrium, which implies that at equilibrium, the framework $\mathcal{F}(\mathcal{G}, p^\dagger, \mathcal{A})$ is strongly distance rigid. Using Proposition 2, we have the equilibrium framework as sign-elevation angle rigid, proving part (ii). $\square$

Theorem 4. *Consider a sign-elevation angle rigid framework constructed by a sign-elevation-Henneberg construction in* 3*-D. Using control law* (10) *for single integrator agents* (1)*, for sufficiently large* κ*, the formation tracking error* e *almost globally converges to zero.*

Proof. We rewrite the control law in (10) as

$$\begin{aligned} \dot{p} &= -(\bar{R}^v_e)^\top(p) e(p) = -R^\top_e(p) e_e(p) - \kappa R^\top_v(p) e_v(p) && (25)\\ &= -R^\top_e(p) e_e(p) - \kappa^2 R^\top_v(p) \bar{e}_v(p), && (26)\end{aligned}$$

where $e_e(p) := E(p) - E^*, e_v(p) := \bar{V}(p) - \bar{V}^*, \bar{e}_v(p) := V(p) - V^*$. Now consider the Lyapunov candidate $\mathcal{L} := \frac{1}{2} e^\top e$, which is positive definite and radially unbounded. Taking its derivative along (10) yields

$$\dot{\mathcal{L}} = e^\top \dot{e} = -e^\top \bar{R}^v_e \bar{R}^{v\top}_e e = -||\bar{R}^{v\top}_e e||^2 \leq 0. \quad (27)$$

From, (27), we have $\dot{\mathcal{L}} \leq 0$. Hence, all the signals are bounded. Also, $\dot{\mathcal{L}} \equiv 0$, implies, $\bar{R}^{v\top}_e e \equiv 0$. Hence, the largest invariant set is given by the solution to $\bar{R}^{v\top}_e e \equiv 0$. Let, there exist non-zero $R^\top_e(p) e_e(p)$, and $R^\top_v(p) e_v(p)$ at equilibrium $p = p^*$ and $\bar{R}^{v\top}_e e = 0$, which implies from (25) that $-R^\top_e(p^*) e_e(p^*) - \kappa^2 R^\top_v(p^*) \bar{e}_v(p^*) = 0$. However, as e is bounded and hence R_e and R_v are also bounded, we may choose sufficiently large $\kappa > 0$ such that $-R^\top_e(p^*) e_e(p^*) - \kappa^2 R^\top_v(p^*) \bar{e}_v(p^*) \neq 0$, which is a contradiction. Hence, at equilibrium, we have

$$R^\top_e(p^*) e_e(p^*) = R^\top_v(p^*) \bar{e}_v(p^*) = 0. \quad (28)$$

From Lemma 3, we already know that for sufficiently large κ we have at equilibrium, the framework $(\mathcal{G}, p)$ being strongly distance rigid, implying rank$(R_e(p^*)) = 3n - 6$. Hence, using the fact that we have a sign-elevation-Henneberg construction and $|\mathcal{E}| = 3n - 6$, $R_e(p^*)$ is of full row rank. Therefore, we have $R^\top_e(p^*) e_e(p^*) = 0$, implying $e_e(p^*) = 0$. Also, from Lemma 3 for sufficiently large κ, note that the signed volume at equilibrium has the same sign as that of the desired volume. Now, as $e_e = 0$, implies the desired elevation angles, and hence, the desired distances are achieved, and the desired volumes are also achieved, i.e., $e_v(p^*) = 0$. Hence, the largest invariant set is given by $e = 0$, hence using

LaSalle's invariance principle [47] we have e goes to zero asymptotically. □

Remark 7. *From Lemma 3 and Theorem 4, there exists a sufficiently large $\kappa > 0$ such that any stable equilibrium p^* of system* (10) *yields the desired formation shape, i.e., $e_e = 0$ and $\bar{e}_v = 0$. Undesired unstable equilibria may exist with $e_e \neq 0$ and $e_v \neq 0$ satisfying $R_e^\top(p^*)e_e(p^*) - \kappa^2 R_v^\top(p^*)\bar{e}_v(p^*) = 0$. Since e_e and $\bar{e}_v$ are interdependent, so are $R_e^\top$ and $R_v^\top$, making this equation highly nonlinear and analytically intractable. However, the set of unstable equilibria is of measure zero and nowhere dense, ensuring the stability result is almost global.*

Remark 8. *While [40] considered only planar frameworks with displacement measurements in the control laws, we extend this setup to 3-D frameworks using bearing-only measurements. The sub-frameworks in [40] generating 2-D frameworks shared no common sides; however, this does not apply to 3-D frameworks. Instead, we use tetrahedral sub-frameworks (Fig. 3) to construct 3-D frameworks. When merging these sub-frameworks, we glue together three sides, since unlike the 2-D case, sub-frameworks with no common sides are not infinitesimally sign-elevation angle rigid in 3-D. For Lemma 3, infinitesimal rigidity of sub-frameworks is essential, so we select those shown in Fig. 3 that satisfy this condition. To merge three sides of adjacent sub-frameworks and form the final framework, we redefine the elevation constraint error as $\hat{e}_{e_{ij}} := \frac{1}{\sqrt{\varrho_{ij}}} e_{e_{ij}}$, where $e_{e_{ij}} := f_{ij} - f_{ij}^*$ and ϱ_{ij} is the number of sub-frameworks sharing edge (i,j). The Hessian in* (19) *is then computed using these errors.*

Remark 9. *Physically, increasing κ adds more weight to the volume error. Our control law is gradient-based, derived from the negative gradient of the potential function Φ. Since the elevation angle and volume errors are interdependent, Φ is a complicated function that may have many local minima. For a low $\kappa > 0$, the system may converge to an equilibrium $p^\dagger \in \mathbb{R}^{nd}$ corresponding to a local minimum of Φ. Increasing κ imposes more weight on the volume error term in* (10)*, altering the shape of Φ and helping the system converge to the desired equilibrium at the global minimum.*

Remark 10. *When disturbances are present in the system dynamics, i.e., $\dot{p}_i = u_i + d_i(t)$ with bounded disturbance $||d_i(t)||_\infty \leq \gamma$, $\forall i \in \mathcal{V}$, the control input u_i can be modified to eliminate their effect. Instead of* (11)*, we use $u_i = -r_c(\bar{R}_e^v)^\top(i,:)(F_e^v - (F_e^v)^*) - k\, sign((\bar{R}_e^v)^\top(i,:)(F_e^v - (F_e^v)^*))$, with $k > \gamma$. For stability, consider the Lyapunov candidate $\mathcal{L} := \frac{1}{2}e^\top e$. Its derivative satisfies $\dot{\mathcal{L}} \leq -||(\bar{R}_e^v)^\top e||^2 - k||(\bar{R}_e^v)^\top e||_1 + \gamma||(\bar{R}_e^v)^\top e||_1 \leq -||(\bar{R}_e^v)^\top e||^2$, since $k > \gamma$. Using similar arguments as in Theorem 1 and Barbalat's lemma [47], asymptotic convergence of errors to zero is established.*

D. Control design for non-holonomic agents

In this subsection, we propose control laws for non-holonomic agents, as described in equation (2), and demonstrate their stability using our proposed laws.

The following control laws are proposed

$$\begin{aligned} \bar{v}_i &= h_i^\top u_i \\ w_i &= h_i \times u_i \; i \in \mathcal{V}_i, \end{aligned} \tag{29}$$

where $u_i \in \mathbb{R}^d$ is the control input proposed for single integrators, i.e., the expression for u_i can be found in (10). Although the controller here has a similar structure as the one in [48], here the control law uses bearing measurements in agents' local frames. On the other hand, the control laws discussed in [48] required inter-agent displacement measurements.

Theorem 5. *Consider a sign-elevation angle rigid framework constructed by a sign-elevation-Henneberg construction in* 2*-D or* 3*-D. Using control law* (29) *for non-holonomic agents in* (2)*, with sufficiently large κ, the formation tracking error, e, almost globally converges to zero.*

Proof. Using the proposed control law in (29), the overall dynamics of n agents becomes

$$\dot{p} = HH^\top u, \quad \dot{h} = H^\perp (H^\perp)^\top u, \tag{30}$$

where $H := \mathrm{diag}(h_1, \ldots, h_n) \in \mathbb{R}^{nd \times nd}, H^\perp := \mathrm{diag}(h_1^\perp, \ldots, h_n^\perp) \in \mathbb{R}^{nd \times nd}$. Now, let the Lyapunov candidate be $\mathcal{L} = \frac{1}{2}e^\top e$, which is positive definite and radially unbounded. The time derivative of $\mathcal{L}$, along the system trajectory (30) is given by

$$\begin{aligned} \dot{\mathcal{L}} &= e^\top \dot{e} = e^\top \bar{R}_e^v \dot{p} = -e^\top \bar{R}_e^v H H^\top (\bar{R}_e^v)^\top e \\ &= -||H^\top u||^2 \leq 0. \end{aligned} \tag{31}$$

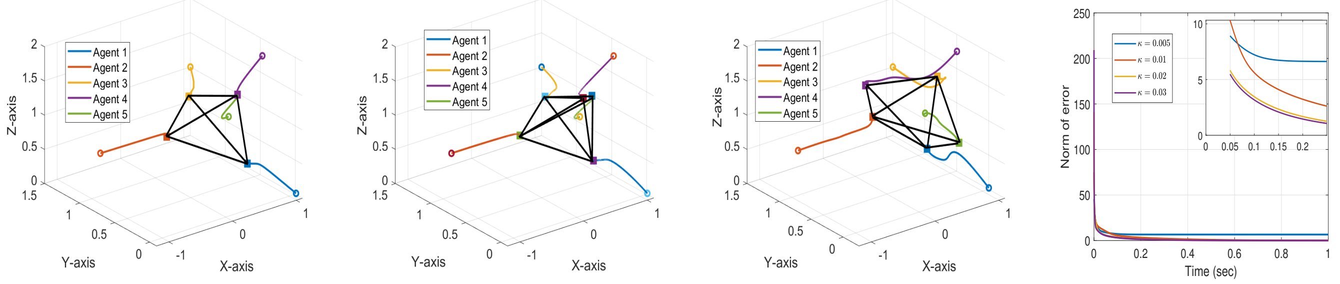
Form (31), it is clear that $\dot{\mathcal{L}}$ is negative semi-definite, and hence, all the signals are bounded. Now, when $\dot{\mathcal{L}} \equiv 0$, either (i) $u_i \equiv 0$, $\forall i \in \mathcal{V}$, or (ii) we have $h_i \perp u_i, u_i \not\equiv 0 \; \forall i \in \mathcal{V}$. Clearly, case (ii) cannot occur because if (ii) occurs, w_i would become non-zero and hence h_i will change and the condition in (ii) will not be satisfied. Hence, we have case (i) as the only possibility, and from Theorem 4 it implies e approaches zero asymptotically. □

IV. Illustrative Examples

A. Single integrators in 3-D

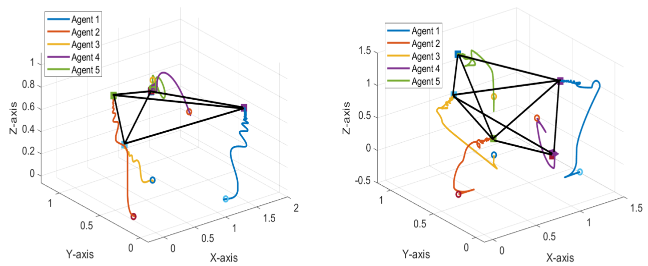
Next, we consider four agents in 3-D, with the edge set $\mathcal{E} = \{(1,2),(2,3),(3,1),(4,1),(4,2),(4,3),(5,1),(5,2),(5,3)\}$ and the volume index set $\mathcal{A} = \{(4,1,2,3),(5,1,2,3)\}$. The desired formation shape is a hexahedron with triangular faces, while the desired elevation angle constraints are given by $f_{ij}^* = 10$, $\forall (i,j) \in \mathcal{E}$, and the desired signed-volume constraints are $V_{4123}^* = -707.1068, V_{5123}^* = 707.1068$. The radius of the ball is $r_c = 0.1$. The initial positions of the agents are $p_1(0) = [1 \; -0.1 \; 0.1,]^\top, p_2(0) = [-1.2 \; 0.7 \; 0.9]^\top, p_3(0) = [1.1 \; 1.5 \; 1]^\top, p_4(0) = [0.8 \; 0.2 \; 2]^\top, p_5(0) = [0.6 \; 0.5 \; 1]^\top$. First, we set $\kappa = 0$ and then the control law becomes identical with [36], and Fig. 4 (a) plots the trajectories of the agents in this case. We may observe that agents 4 and 5 coincide at one position, giving rise to a final shape that is not congruent to the desired formation shape. With $\kappa = 0.005$ we have Fig. 4 (b), where also the desired formation shape is not achieved due to very small value of κ. Now, we set $\kappa = 0.02$ and the

This article has been accepted for publication in IEEE Transactions on Control of Network Systems. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/TCNS.2025.3638076



(a) Single integrators for $\kappa = 0$ (using [36]) (b) Single integrators for $\kappa = 0.005$ (c) Single integrators for $\kappa = 0.02$ (d) Norm of error for different κ

Fig. 4. Trajectories of single integrators in 3-D for (a) $\kappa = 0$, (b) $\kappa = 0.005$, (c) $\kappa = 0.02$, (d) Norm of error $\left(||e|| \text{ where } e = [e_e^\top \; e_v^\top]^\top\right)$ for different κ



(a) Non-holonomic agents for $\kappa = 0$ (b) Non-holonomic agents for $\kappa = 0.02$

Fig. 5. Trajectories of non-holonomic agents for (a) $\kappa = 0$, (b) $\kappa = 0.02$.

plot for this case is given in Fig. 4 (c), which indicates that the desired shape is achieved with our proposed law. In Fig. 4 (d), the norm of total error, i.e., $||e||$, is plotted for different values of κ. Fig. 4 (d) shows that while for very small values of κ, i.e., $\kappa = 0.005$, the errors do not converge to zero, for higher values of κ such as $\kappa = 0.01, \kappa = 0.02$, or $\kappa = 0.04$ the error plot does converge to zero. Also, the convergence rate increases with higher values of κ.

B. Non-holonomic agents

For non-holonomic agents, we consider the same desired formation shape as for the case of single integrators in 3-D, with the same desired elevation angle and signed-volume constraints. The initial position of the agents are $p_1(0) = [1 \;\; -0.1 \;\; 0.1]^\top, p_2(0) = [-0.1 \;\; 0.2 \;\; 0.02]^\top, p_3(0) = [0.6 \;\; 0.5 \;\; 0.1]^\top, p_4(0) = [0.8 \;\; 0.2 \;\; 0.8]^\top$, and $p_5(0) = [0.6 \;\; 0.5 \;\; 1]^\top$, and the initial headings of the agents are $h_1(0) = [1 \;\; 0 \;\; 0]^\top, h_2(0) = [-1 \;\; 0 \;\; 0]^\top, h_3(0) = [1/\sqrt{2} \;\; 1/\sqrt{2} \;\; 0]^\top, h_4(0) = [0 \;\; 1/\sqrt{3} \;\; \sqrt{2}/\sqrt{3}]^\top, h_5(0) = [1/\sqrt{3} \; 1/\sqrt{3} \; 1/\sqrt{3}]^\top$ and $h_6(0) = [\frac{1}{\sqrt{3}} \; \frac{\sqrt{2}}{\sqrt{3}}]^\top$. Fig. 5 (a) shows the trajectories for the non-holonomic agents with $\kappa = 0$, and the agents are converging to an ambiguous shape where agent 4 and 5 are converging to the same position. Fig. 5 (b) shows the trajectories for the non-holonomic agents with our proposed control law (29) with $\kappa = 0.002$, and the desired formation shape is achieved.

V. CONCLUSIONS

In this work, we have proposed sign-elevation angle rigidity for uniquely characterizing formation shapes using elevation angle signed-volume constraints for 3-D graphs, respectively. Then, bearing-only control laws are derived using only agents' local bearing information. Compared with the elevation angle rigidity-based control laws in [36], [38], which can only ensure local stability results, we have proved almost global results for the frameworks generated by sign-elevation-Henneberg construction, in which the overall framework consists of tetrahedral sub-frameworks for 3-D graphs. A gain parameter κ was introduced for signed-volume constraints, and it was proved that there always exists a sufficiently large κ such that agents can achieve the desired formation shape almost globally. Determining the exact lower bound for κ is a future topic for research. The results presented here can be extended for obtaining bearing-only formation maneuvers using similar maneuvering methods as in [36].

APPENDIX A

- **Proof of Lemma 2:**

Define a function $\zeta_i(\beta_i)|_{\mathcal{G}_j}$ near $(\mathcal{G}_j, \mathcal{A}_j, \beta_j)$ as $\zeta_i(\beta_i)|_{\mathcal{G}_j} = \beta_i \in \mathcal{B}_{\beta_j}$ which can be written as

$$\zeta_i(\beta_i)|_{\mathcal{G}_j} = \begin{cases} \zeta_i \in \mathbb{R}^{3|\mathcal{V}_i|} & \text{if } i = j, \\ \zeta_i \in \mathcal{B}_{\beta_j} \subseteq \mathbb{R}^{3|\mathcal{V}_i|} & i \neq j, \end{cases} \tag{32}$$

which implies for the case $i = j$ as the β_i is an interior point of $\mathcal{B}_{\beta_i}$ the sub-framework β_i completely belongs to $\mathcal{B}_{\beta_i}$. Now, as β_j is a regular point of $F^v_{e_j}$ using [8, Proposition 2], there exists a neighborhood $\mathcal{B}_{\beta_j}$ such that $(F^v_{e_j})^{-1} F^v_{e_j}(\beta_j) \cap \mathcal{B}_{\beta_j}$ is a $3|\mathcal{V}_j| - r^m_{F^v_{e_j}}$, where $r^m_{F^v_{e_j}} = \max \left\{ \text{rank} \left(\frac{\partial F^v_{e_j}}{\partial \beta_j} \right) \middle| \beta_j \in \mathbb{R}^{3|\mathcal{V}_j|} \right\}$. Hence, for the case when $i \neq j$ there exists $\zeta_i(\beta_i)|_{\mathcal{G}_j}$ such that $\zeta_i(\beta_i)|_{\mathcal{G}_j} = \beta_i \in (F^v_{e_j})^{-1} F^v_{e_j}(\beta_j)$. Hence we have

$$\zeta_i(\beta_i)|_{\mathcal{G}_j} = \begin{cases} \zeta_i \in \mathbb{R}^{3|\mathcal{V}_i|} & \text{if } i = j, \\ \zeta_i \in \mathcal{B}_{\beta_j} \subseteq (F^v_{e_j})^{-1} F^v_{e_j}(\beta_j), \; i \neq j. \end{cases} \tag{33}$$

Hence we have a derivative map $D\beta_i$ for $i \neq j$ such that $D\zeta_i(\beta_i)|_{\mathcal{G}_j} = v_i \in T_{\zeta_i(\beta_i)}(F^v_{e_j})^{-1}F^v_{e_j}(\beta_j)$, which implies

$$D\zeta_i(v_i)|_{\mathcal{G}_j} = \begin{cases} v_i \in \mathbb{R}^{3|\mathcal{V}_i|} & \text{if } i = j, \\ v_i \in \mathcal{B}_{\beta_j} \subseteq T_{\zeta_i(\beta_i)}(F^v_{e_j})^{-1}F^v_{e_j}(\beta_j), & i \neq j. \end{cases} \tag{34}$$

This completes the proof.

Appendix B

• **Expression of Hessian for 3-D sub-frameworks :**

Here, we calculate the Hessian associated with the sub-framework in the case of 3-D, which is given by

$$\mathcal{H}_{\beta_w} = \frac{\partial^2 \Phi_w}{\partial \beta_w^2} = \frac{\partial}{\partial \beta_w}(\bar{R}^{v\top}_{e_w} K^2 e_w) \in \mathbb{R}^{12\times 12}, \tag{35}$$

where $e_w(\beta_w) = [E_w^\top(\beta_w)\ V_w(\beta_w)]^\top - [E_w^{*\top}\ V_w^*]^\top, \Phi_w = \frac{1}{2}||\hat{e}_w(\beta_w)||^2$, and $\beta_w = [p_i^\top\ p_j^\top\ p_k^\top\ p_l^\top]^\top$. Consider $w \in \{1, \ldots, \psi\}$, hence $F^v_e = [f_{ij}\ f_{ik}\ f_{il}\ f_{kj}\ f_{kl}\ f_{jl}\ \kappa V_{ijkl}]^\top$ (see Fig. 3). The sign-elevation angle rigidity matrix of the sub-framework is calculated as:

$$\bar{R}^v_{e_w} = \frac{\partial F^v_{e_w}}{\partial \beta_w},$$

which is given by

$$\bar{R}^v_{e_w} = \frac{1}{r_c}\begin{bmatrix} -b_{ij}^\top & b_{ij}^\top & 0 & 0 \\ -b_{ik}^\top & 0 & b_{ik}^\top & 0 \\ -b_{il}^\top & 0 & 0 & b_{il}^\top \\ 0 & b_{kj}^\top & -b_{kj} & 0 \\ 0 & 0 & -b_{kl}^\top & b_{kl}^\top \\ 0 & -b_{jl}^\top & 0 & b_{jl}^\top \\ r_c^{-2}\kappa Z_1^\top & r_c^{-2}\kappa Z_2^\top & r_c^{-2}\kappa Z_3^\top & r_c^{-2}\kappa Z_4^\top \end{bmatrix},$$

where $Z_1 = z_{kl} \times z_{kj}, Z_2 = z_{ik} \times z_{il}, Z_3 = z_{il} \times z_{ij}, Z_4 = z_{ij} \times z_{ik}$. Note that $Z_1 = -Z_2 - Z_3 - Z_4$. Now, the Hessian is calculated as

$$\mathcal{H}_{\beta_w} = \frac{\partial}{\partial \beta_w}(\bar{R}^{v\top}_{e_w} K^2 e_w) = \bar{N}_1 + \bar{N}_2 + (\bar{R}^v_{e_w})^\top K^4 \bar{R}^v_{e_w},$$

where $\bar{N}_2 = \frac{\kappa e_{v_{ijkl}}}{r_c^3} N_2 \in \mathbb{R}^{12\times 12}$, $N_2 =$

$$\begin{bmatrix} 0 & -[z_{kl}]^\times & [z_{kl}]^\times - [z_{kj}]^\times & [z_{kj}]^\times \\ [z_{ik}]^\times - [z_{il}]^\times & 0 & [z_{il}]^\times & -[z_{ik}]^\times \\ [z_{il}]^\times - [z_{ij}]^\times & -[z_{kl}]^\times & 0 & [z_{ij}]^\times \\ [z_{ij}]^\times - [z_{ik}]^\times & [z_{ik}]^\times & -[z_{ij}]^\times & 0 \end{bmatrix},$$

and $\bar{N}_1 =$

$$r_c^{-1}\begin{bmatrix} Y_1 & -\frac{P_{b_{ij}}}{||z_{ij}||}\tilde{e}_{e_{ij}} & -\frac{P_{b_{ik}}}{||z_{ik}||}\tilde{e}_{e_{ik}} & -\frac{P_{b_{il}}}{||z_{il}||}\tilde{e}_{e_{il}} \\ -\frac{P_{b_{ij}}}{||z_{ij}||}\tilde{e}_{e_{ij}} & Y_2 & -\frac{P_{b_{kj}}}{||z_{kj}||}\tilde{e}_{e_{kj}} & -\frac{P_{b_{jl}}}{||z_{jl}||}\tilde{e}_{e_{jl}} \\ -\frac{P_{b_{ik}}}{||z_{ik}||}\tilde{e}_{e_{ik}} & -\frac{P_{b_{kj}}}{||z_{kj}||}\tilde{e}_{e_{kj}} & Y_3 & -\frac{P_{b_{kl}}}{||z_{kl}||}\tilde{e}_{e_{kl}} \\ -\frac{P_{b_{il}}}{||z_{il}||}\tilde{e}_{e_{il}} & -\frac{P_{b_{jl}}}{||z_{jl}||}\tilde{e}_{e_{jl}} & -\frac{P_{b_{kl}}}{||z_{kl}||}\tilde{e}_{e_{kl}} & Y_4 \end{bmatrix},$$

where $Y_1 = \frac{P_{b_{ij}}}{||z_{ij}||}\tilde{e}_{e_{ij}} + \frac{P_{b_{ik}}}{||z_{ik}||}\tilde{e}_{e_{ik}} + \frac{P_{b_{il}}}{||z_{il}||}\tilde{e}_{e_{il}}, Y_2 = \frac{P_{b_{ij}}}{||z_{ij}||}\tilde{e}_{e_{ij}} + \frac{P_{b_{kj}}}{||z_{kj}||}\tilde{e}_{e_{kj}} + \frac{P_{b_{jl}}}{||z_{jl}||}\tilde{e}_{e_{jl}}, Y_3 = \frac{P_{b_{ik}}}{||z_{ik}||}\tilde{e}_{e_{ik}} + \frac{P_{b_{kj}}}{||z_{kj}||}\tilde{e}_{e_{kj}} + \frac{P_{b_{kl}}}{||z_{kl}||}\tilde{e}_{e_{kl}}, Y_4 = \frac{P_{b_{il}}}{||z_{il}||}\tilde{e}_{e_{il}} + \frac{P_{b_{jl}}}{||z_{jl}||}\tilde{e}_{e_{jl}} + \frac{P_{b_{kl}}}{||z_{kl}||}\tilde{e}_{e_{kl}}$, with $P_{b_{ij}} := I_3 - b_{ij}b_{ij}^\top \in \mathbb{R}^{3\times 3}$ is the projection matrix, and $\tilde{e}_{e_{ij}} := \frac{1}{\varrho_{ij}} e_{e_{ij}}, (i,j) \in \mathcal{E}_w$.

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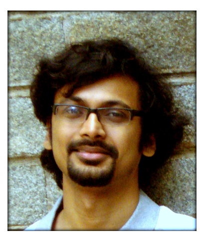

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